

Frame-by-frame forward chaining on the Mechanism KG

r_1 : $\text{hasInput}(m, q) \leftarrow \text{hasApplication}(m, a) \wedge \text{differentiand}(a, q)$
 r_2 : $\text{hasRefVar}(m, v) \leftarrow \text{hasApplication}(m, a) \wedge \text{withRespectTo}(a, v)$
 r_3 : $\text{type}(m, \text{RateOfChange}) \leftarrow \text{type}(m, \text{Mech}) \wedge \text{hasInput} \wedge \text{hasRefVar}$

Frame 0 — G_0 (asserted)

rateOfChange_1 hasApplication $\text{derivativeApplication_1}$
 $\text{derivativeApplication_1}$ differentiand position_1
 $\text{derivativeApplication_1}$ withRespectTo time_1

r_1, r_2 fire
 $\Rightarrow$ 2 new

Frame 1 — $G_1 = T_P(G_0) \cup G_0$

...(the three triples of G_0)

rateOfChange_1 hasInput position_1 (+new, r_1)

rateOfChange_1 $\text{hasReferenceVariable}$ time_1 (+new, r_2)

r_3 fires
 $\Rightarrow$ 1 new

Frame 2 — $G_2 = T_P(G_1) \cup G_1$

...(the five triples of G_1)

rateOfChange_1 rdf:type $\text{RateOfChangeMechanism}$ (+new, r_3)

no new
 $\Rightarrow$ halt

Frame 3 — $G_3 = T_P(G_2) \cup G_2 = G_2$ (fixpoint)

One pass over the rules produces 0 new triples $\Rightarrow$ the system halts.

The closure $\text{lfp}(T_P)$ is materialized.